

HOUSTON BUSH, TX, USA

WMO: 722430

Lat: 29.980N

Lon: 95.360W

Elev: 29

StdP: 100.98

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	1.3	-8.7	1.8	5.1	-6.1	2.3	7.5	10.6	13.4	9.2	12.1	2.6	350	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	36.4	24.8	35.4	24.9	34.4	24.9	26.8	31.6	26.4	31.2	26.1	30.9	3.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.0	28.3	25.3	20.5	28.0	25.0	20.1	27.9	83.7	31.5	82.1	31.3	80.8	30.9	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.2	7.4	DB	-3.2	38.7	2.0	1.8	-4.6	39.9	-5.8	40.9	-7.0	41.9	-8.5	43.2
			WB	-4.6	27.7	1.9	0.6	-6.0	28.1	-7.1	28.5	-8.1	28.8	-9.5	29.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	12.0	14.1	17.4	21.0	25.1	28.1	29.1	29.3	26.7	22.0	16.7	12.8	(6)
(7)		DBStd	7.33	5.37	5.40	4.67	3.71	2.78	1.93	1.57	1.82	2.56	3.98	4.83	5.30	(7)
(8)		HDD10.0	101	41	21	5	0	0	0	0	0	0	0	6	29	(8)
(9)		HDD18.3	721	206	136	74	18	1	0	0	0	0	14	86	186	(9)
(10)		CDD10.0	4200	102	136	234	329	468	542	592	598	502	372	208	117	(10)
(11)		CDD18.3	1778	9	19	46	97	210	292	334	339	252	128	37	16	(11)
(12)		CDH23.3	18091	28	78	222	661	1936	3272	4003	4039	2482	1081	232	59	(12)
(13)	CDH26.7	7862	0	8	27	140	698	1514	1967	2047	1083	343	32	3	(13)	
(14)	Wind	WSAvg	3.3	3.5	3.7	3.8	4.0	3.7	3.0	2.6	2.6	2.7	3.0	3.2	3.3	(14)
(15)	Precipitation	PrecAvg	1267	90	74	89	92	133	139	95	112	114	133	98	95	(15)
(16)		PrecMax	2026	248	155	216	366	366	488	328	993	313	408	298	237	(16)
(17)		PrecMin	582	9	10	3	3	1	2	8	2	20	1	10	16	(17)
(18)		PrecStd	299	55	45	53	77	95	118	66	142	76	105	65	48	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.7	27.8	29.1	31.3	34.5	36.9	37.4	38.2	36.8	33.0	29.3	26.9	(19)
(20)			MCWB	19.1	19.3	20.1	22.0	23.5	24.3	24.5	24.7	23.9	23.7	22.5	21.2	(20)
(21)		2%	DB	24.0	25.6	27.3	29.5	32.8	35.1	36.0	36.6	34.6	31.6	27.6	25.0	(21)
(22)			MCWB	18.4	19.4	20.1	21.4	23.6	24.6	24.9	25.1	24.1	23.6	21.4	20.3	(22)
(23)		5%	DB	22.5	23.9	25.9	28.3	31.6	34.0	35.0	35.5	33.3	30.2	26.0	23.4	(23)
(24)			MCWB	18.5	18.9	19.6	21.1	23.7	24.7	25.0	25.2	24.2	22.8	20.7	20.1	(24)
(25)		10%	DB	21.0	22.3	24.3	27.0	30.4	33.0	33.9	34.3	32.1	28.6	24.4	21.9	(25)
(26)			MCWB	17.7	18.6	19.0	20.5	23.4	24.7	25.1	25.2	24.0	21.9	20.3	19.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	21.8	22.6	24.5	26.3	27.4	27.1	27.1	26.8	26.2	24.3	22.7	(27)
(28)			MCDWB	22.9	24.4	26.0	28.1	31.1	31.5	32.1	32.7	30.6	29.0	26.8	24.5	(28)
(29)		2%	WB	20.4	21.0	21.8	23.6	25.5	26.6	26.7	26.7	26.2	25.4	23.2	21.7	(29)
(30)			MCDWB	22.5	23.6	24.8	26.8	29.9	30.9	31.6	32.2	30.3	28.7	25.8	23.8	(30)
(31)		5%	WB	19.6	20.4	21.1	22.9	24.9	26.1	26.3	26.3	25.7	24.7	22.2	20.9	(31)
(32)			MCDWB	21.9	22.8	23.9	26.0	29.1	30.5	31.2	31.7	29.8	27.9	24.8	23.2	(32)
(33)		10%	WB	18.5	19.4	20.5	22.3	24.4	25.6	25.9	26.0	25.3	23.9	21.1	19.6	(33)
(34)			MCDWB	20.7	21.8	23.2	25.4	28.4	30.1	30.7	31.2	29.4	27.0	23.7	21.5	(34)
(35)	Mean Daily Temperature Range	MDBR	10.7	10.5	10.6	10.5	9.9	9.6	9.8	10.1	10.2	11.1	11.0	10.4	(35)	
(36)		5% DB	MCDBR	11.6	10.9	10.6	10.4	10.4	10.8	11.0	11.5	11.5	11.3	11.5	11.1	(36)
(37)			MCWBR	7.4	6.1	5.2	4.5	3.5	2.8	2.7	2.8	3.5	4.4	6.2	7.2	(37)
(38)		5% WB	MCDBR	9.7	9.1	9.0	8.3	8.9	9.0	9.8	10.1	9.1	9.0	9.8	9.8	(38)
(39)			MCWBR	7.3	6.4	5.8	4.5	3.7	3.1	3.0	2.9	3.3	4.6	6.6	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.360	0.374	0.436	0.482	0.471	0.498	0.484	0.454	0.380	0.372	0.347	(40)	
(41)		taud	2.502	2.454	2.406	2.233	2.143	2.223	2.172	2.213	2.258	2.454	2.425	2.487	(41)	
(42)		Ebn at Noon	895	904	911	862	820	824	801	810	822	870	847	866	(42)	
(43)		Edh at Noon	90	103	115	141	155	143	150	142	132	101	97	88	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.87	3.22	4.41	5.30	5.87	6.13	6.06	5.79	4.99	4.30	3.23	2.53	(44)	
(45)		RadStd	0.27	0.57	0.39	0.36	0.43	0.50	0.53	0.46	0.50	0.54	0.33	0.31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+91	+76
(47)	Regional (36 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+97

Nomenclature: See separate page